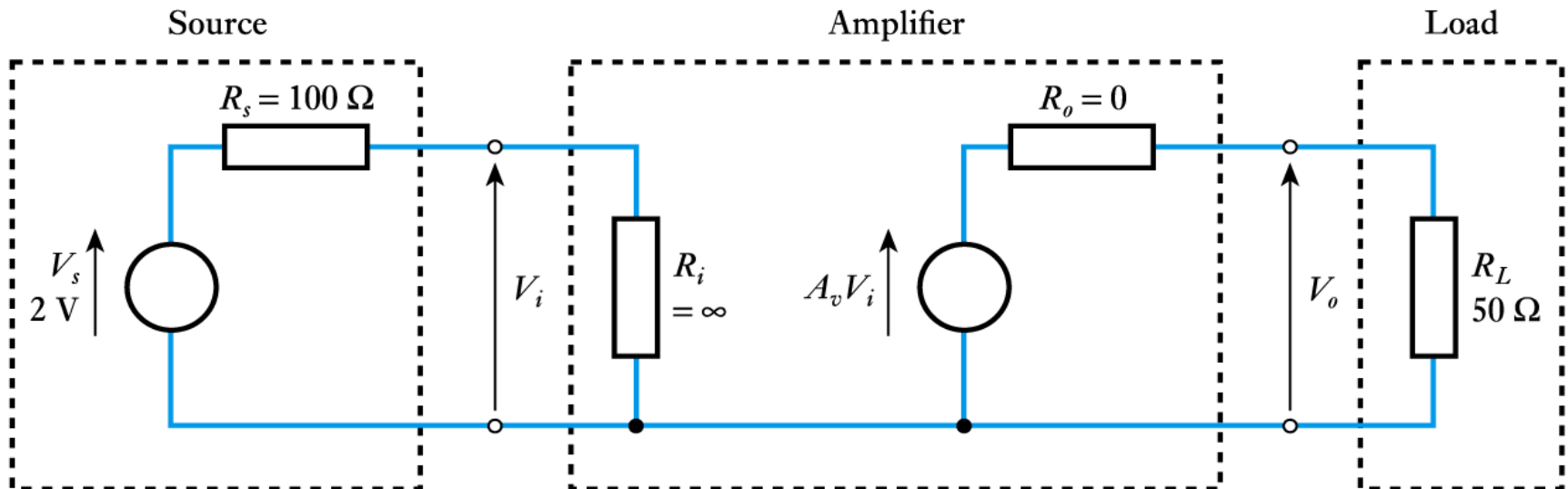


Amplification

- Introduction
- Electronic amplifiers
- Sources and loads
- Equivalent circuit of an amplifier
- **Thévenin and Norton equivalent circuits**
- **Output power**
- **Power gain**
- **Frequency response and bandwidth**
- **Low- and high-frequency effects**
- Noise
- Differential amplifiers
- Simple amplifiers.

Equivalent circuits of amplifiers (contd.)

- An **ideal voltage amplifier** would not suffer from loading
 - it would have $R_i = \infty$ and $R_o = 0$
 - consider the effect on the previous example.



Thévenin and Norton equivalent circuits

■ Thévenin's Theorem

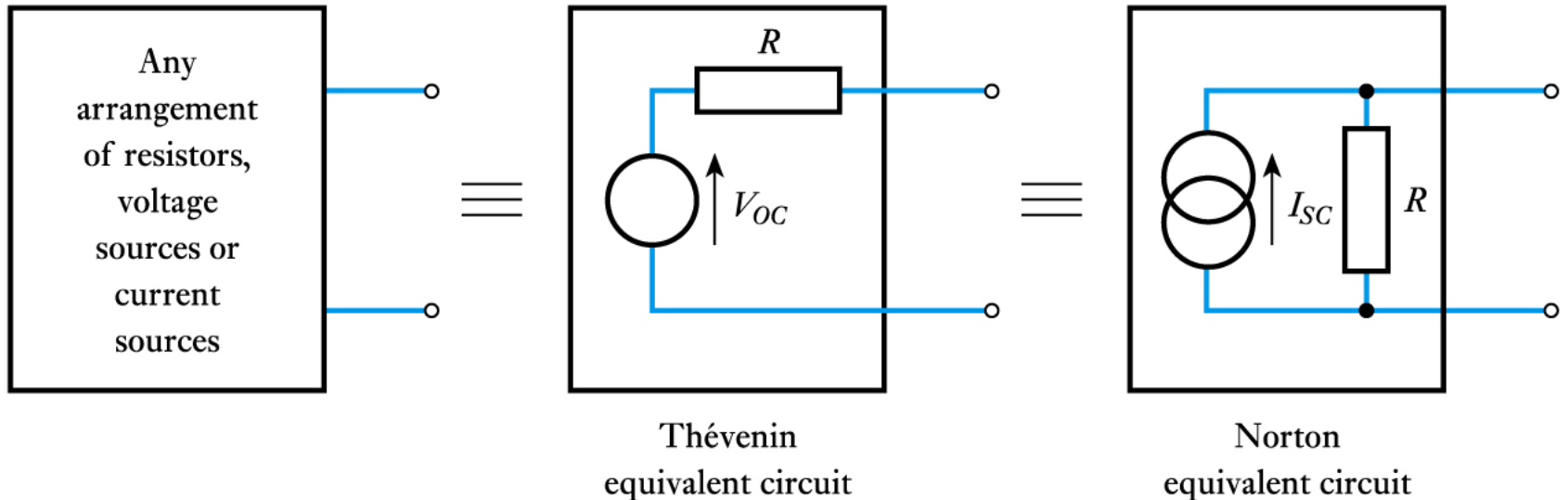
As far as its appearance from outside is concerned, any two terminal network of resistors and energy sources can be replaced by a series combination of an ideal voltage source V and a resistor R , where V is the open-circuit voltage of the network and R is the voltage that would be measured between the output terminals if the energy sources were removed and replaced by their internal resistance.

Thévenin and Norton equivalent circuits (contd.)

■ Norton's Theorem

As far as its appearance from outside is concerned, any two terminal network of resistors and energy sources can be replaced by a parallel combination of an ideal current source I and a resistor R , where I is the short-circuit current of the network and R is the voltage that would be measured between the output terminals if the energy sources were removed and replaced by their internal resistance.

Thévenin and Norton equivalent circuits (contd.)



– from the Thévenin equivalent circuit

$$I_{SC} = \frac{V_{OC}}{R}$$

– hence for *either* circuit

$$R = \frac{V_{OC}}{I_{SC}}$$

Output power

- The output power P_o is that dissipated in the load resistor

$$P_o = \frac{V_o^2}{R_L}$$

- Power transfer is at a *maximum* when $R_L = R_o$
 - **maximum power theorem**
 - choosing a load to maximize power transfer is called **matching**
 - often voltage gain is more important than power transfer.

Power gain

- Power gain is the ratio of the power supplied to the load to that absorbed at the input.

$$P_i = \frac{V_i^2}{R_i}$$

$$P_o = \frac{V_o^2}{R_L}$$

- For numerical example see **Example 3.6** in set text.
- Gain often given in decibels.

$$\text{Power gain (dB)} = 10 \log_{10} \frac{P_2}{P_1}$$

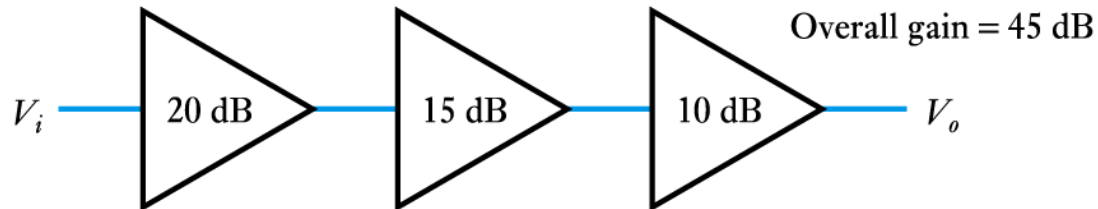
Power gain (contd.)

- Sample gains expressed in dBs

Power gain	Decibels (dBs)
100	20
10	10
1	0

Power gain	Decibels (dBs)
0.5	-3
0.1	-10
0.01	-20

- Using dBs simplifies calculation in cascaded circuits



Power gain (contd.)

- Power gain is related to voltage gain

$$\text{Power gain (dB)} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_2}{V_1^2/R_1}$$

- If $R_1 = R_2$

$$\text{Power gain (dB)} = 10 \log_{10} \frac{V_2^2}{V_1^2} = 20 \log_{10} \frac{V_2}{V_1}$$

$$\text{Power gain (dB)} = 20 \log_{10} (\text{Voltage gain})$$

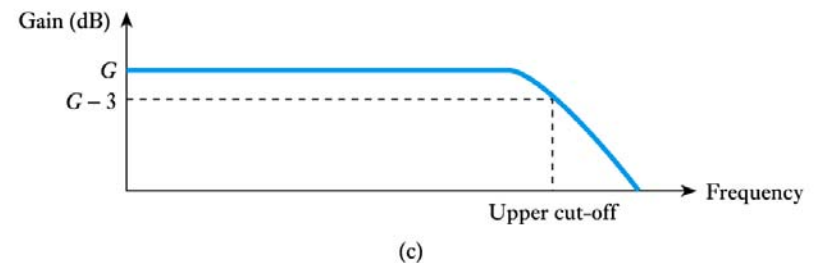
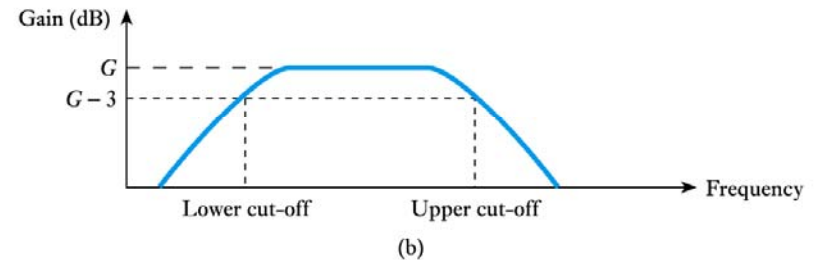
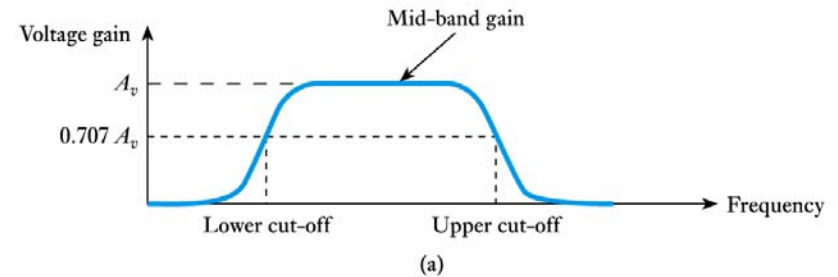
- This expression is often used even when $R_1 \neq R_2$
 - see **Example 3.8** and **Example 3.9** in the course text.

Frequency response and bandwidth

- All real amplifiers have limits to the range of frequencies over which they can be used.
- The gain of a circuit in its normal operating range is termed its **mid-band gain**.
- The gain of all amplifiers falls at high frequencies
 - characteristic defined by the **half-power point**
 - gain falls to $1/\sqrt{2} = 0.707$ times the mid-band gain
 - this occurs at the **cut-off frequency**.
- In some amplifiers gain also falls at low frequencies
 - these are **AC coupled amplifiers**.

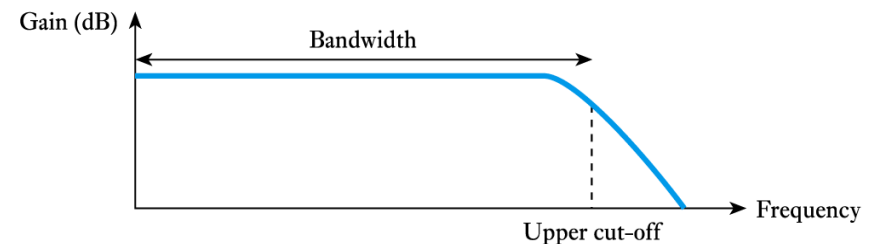
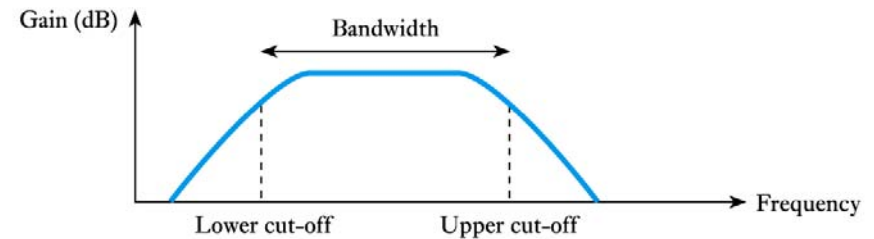
Frequency response and bandwidth (contd.)

- (a) shows an AC coupled amplifier.
- (b) shows the same amplifier – with gain in dBs.
- (c) shows a DC coupled amplifier – the gain is constant down to DC.



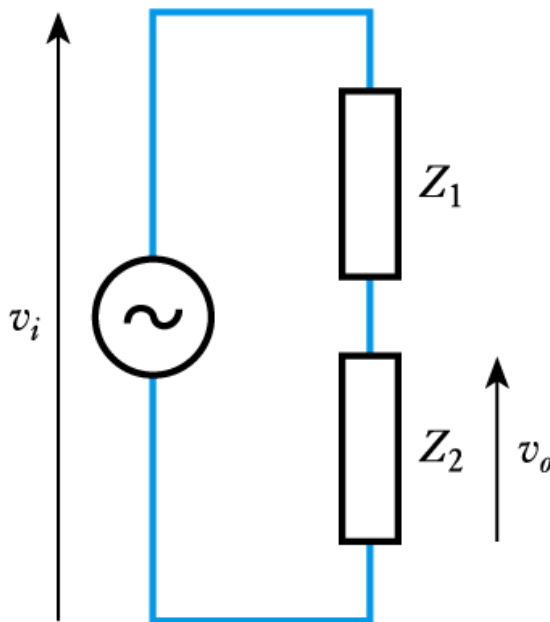
Frequency response and bandwidth (contd.)

- The **bandwidth** is the difference between the upper and lower cut-off frequencies ...
- ... or the difference between the upper-cut-off frequency and zero in a DC coupled amplifier.



Low- and High-frequency effects

- Consider the potential divider shown here



- the output voltage is given by

$$V_o = V_i \times \frac{Z_2}{Z_1 + Z_2}$$

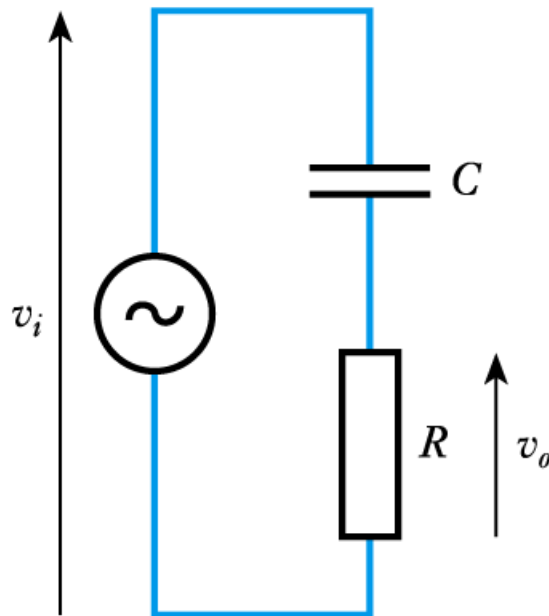
- rearranging, the gain of the circuit is

$$\frac{V_o}{V_i} = \frac{Z_2}{Z_1 + Z_2}$$

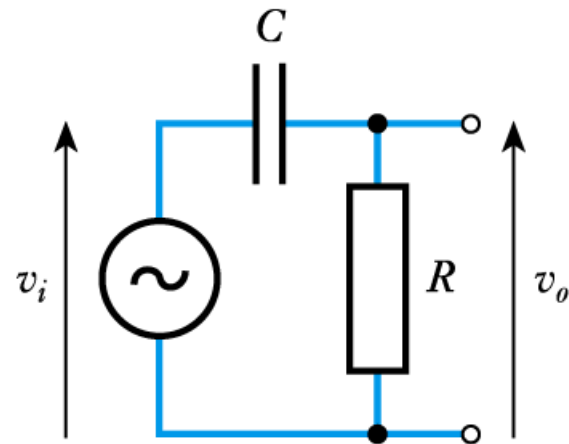
- this is also called the **transfer function** of the circuit

A high-pass RC network

- Consider the following circuit
 - which is shown re-drawn in a more usual form.



(a)



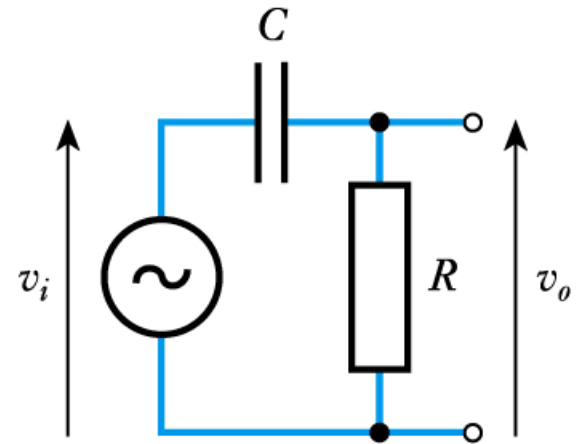
(b)

A high-pass RC network (contd.)

- Clearly the transfer function is

$$\frac{V_o}{V_i} = \frac{Z_R}{Z_R + Z_C} = \frac{R}{R - j\frac{1}{\omega C}} = \frac{1}{1 - j\frac{1}{\omega CR}}$$

- At high frequencies
 - ω is large, voltage gain ≈ 1
- At low frequencies
 - ω is small, voltage gain $\rightarrow 0$



A high-pass RC network (contd.)

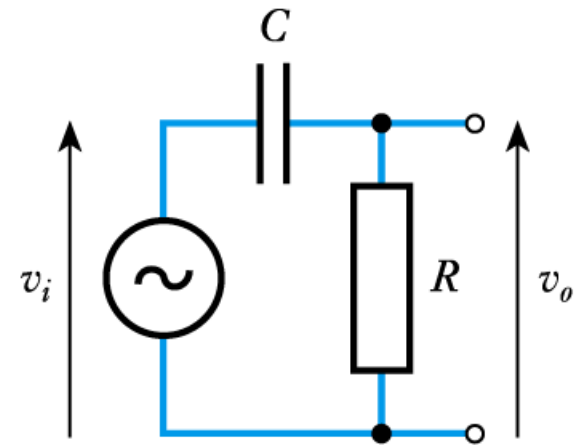
- Since the denominator has real and imaginary parts, the *magnitude* of the voltage gain is

$$|\text{Voltage gain}| = \frac{1}{\sqrt{1^2 + \left(\frac{1}{\omega CR}\right)^2}}$$

- When $1/\omega CR = 1$

$$|\text{Voltage gain}| = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} = 0.707$$

- This is a halving of power, or a fall in gain of 3 dB.



A high-pass *RC* network (contd.)

- The half-power point is the **cut-off frequency** of the circuit.
 - the angular frequency ω_c at which this occurs is given by

$$\frac{1}{\omega_c CR} = 1$$

$$\omega_c = \frac{1}{CR} = \frac{1}{T} \text{ rad/s}$$

- where T is the time constant of the CR network. Also

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi CR} \text{ Hz}$$

A high-pass RC network (contd.)

- Substituting $\omega = 2\pi f$ and $CR = 1/2\pi f_c$ in the earlier equation gives

$$\frac{V_o}{V_i} = \frac{1}{1 - j\frac{1}{\omega CR}} = \frac{1}{1 - j\frac{1}{(2\pi f)\left(\frac{1}{2\pi f_c}\right)}} = \frac{1}{1 - j\frac{f_c}{f}}$$

- This is the general form of the gain of the circuit.
- It is clear that both the *magnitude* of the gain and the *phase angle* vary with frequency.

A high-pass RC network (contd.)

- Consider the behaviour of the circuit at different frequencies:

- **when $f \gg f_c$**

– $f_c/f \ll 1$, the voltage gain ≈ 1

- **when $f = f_c$**

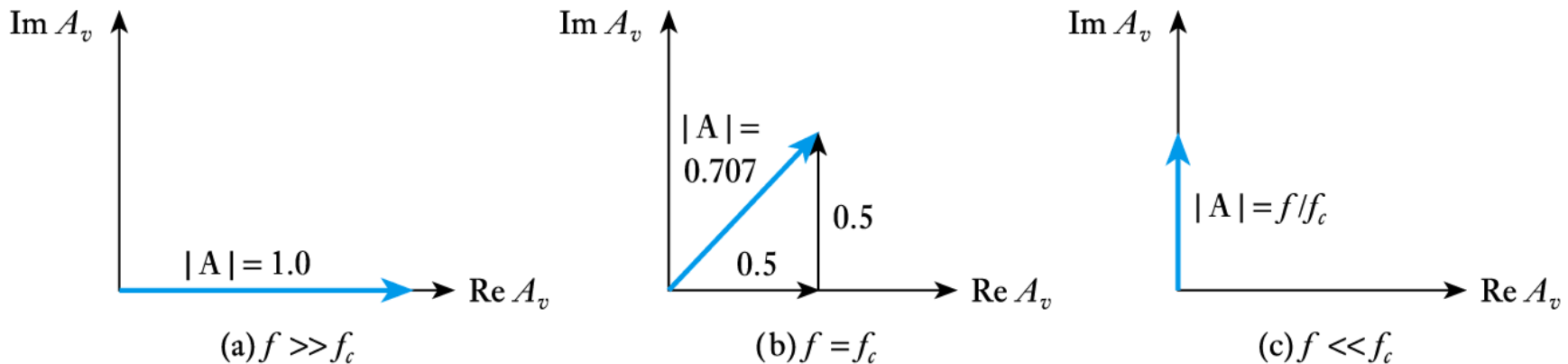
$$\frac{V_o}{V_i} = \frac{1}{1 - j\frac{f_c}{f}} = \frac{1}{1 - j} = \frac{1 \times (1 + j)}{(1 - j) \times (1 + j)} = \frac{(1 + j)}{2} = 0.5 + 0.5j$$

- **when $f \ll f_c$**

$$\frac{V_o}{V_i} = \frac{1}{1 - j\frac{f_c}{f}} \approx \frac{1}{-j\frac{f_c}{f}} = j\frac{f}{f_c}$$

A high-pass RC network (contd.)

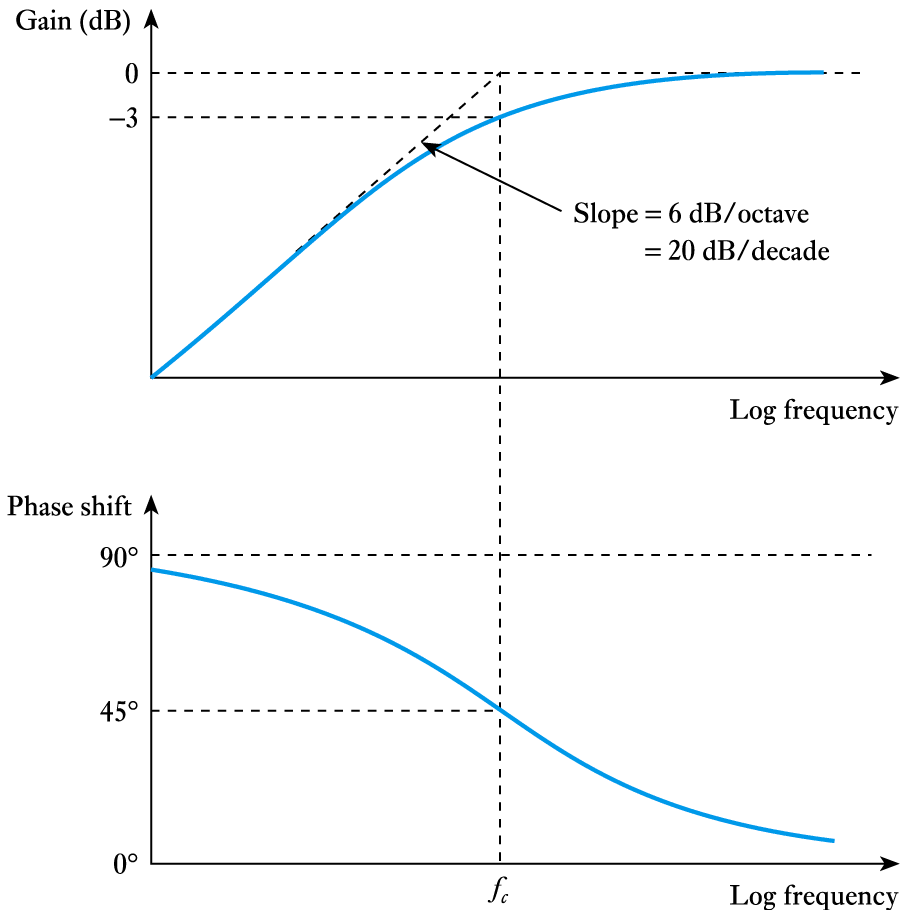
- The behaviour in these three regions can be illustrated using phasor diagrams.



- At *low* frequencies the gain is linearly related to frequency. It falls at -6dB/octave (-20dB/decade).

A high-pass *RC* network (contd.)

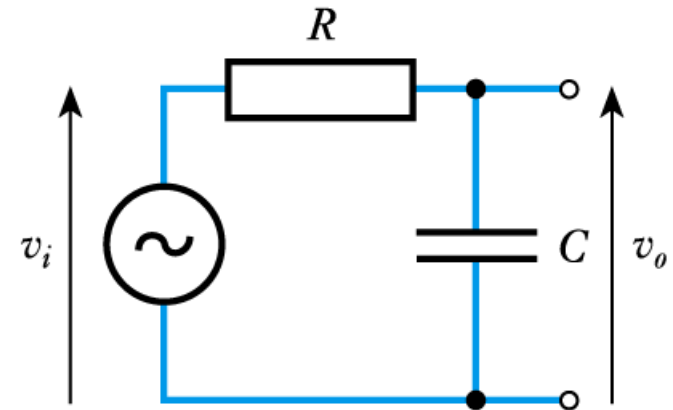
- Frequency response of the high-pass network
 - the gain response has two *asymptotes* that meet at the cut-off frequency
 - figures of this form are called **Bode diagrams**.



A low-pass RC network

- Transposing the C and R gives

$$\frac{V_o}{V_i} = \frac{Z_C}{Z_R + Z_C} = \frac{-j\frac{1}{\omega C}}{R - j\frac{1}{\omega C}} = \frac{1}{1 + j\omega CR}$$



- At high frequencies
 - ω is large, voltage gain $\rightarrow 0$
- At low frequencies
 - ω is small, voltage gain ≈ 1

A low-pass RC network (contd.)

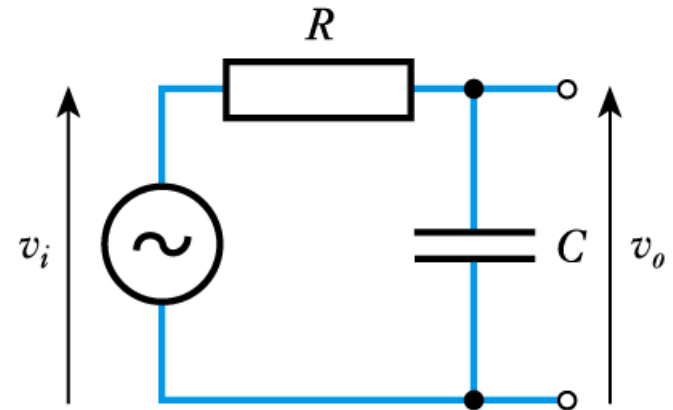
- A similar analysis to that in last section gives

$$|\text{Voltage gain}| = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

- therefore, when $\omega CR = 1$

$$|\text{Voltage gain}| = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} = 0.707$$

- which is the cut-off frequency.



A low-pass *RC* network (contd.)

- Therefore

- the angular frequency ω_c at which this occurs is given by

$$\omega_c CR = 1$$

$$\omega_c = \frac{1}{CR} = \frac{1}{T} \text{ rad/s}$$

- where T is the time constant of the CR network, and as before

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi CR} \text{ Hz}$$

A low-pass RC network (contd.)

- Substituting $\omega = 2\pi f$ and $CR = 1/2\pi f_c$ in the earlier equation gives

$$\frac{V_o}{V_i} = \frac{1}{1 + j\omega CR} = \frac{1}{1 + j\frac{\omega}{\omega_c}} = \frac{1}{1 + j\frac{f}{f_c}}$$

- This is similar, but not the same, as the transfer function for the high-pass network.

A low-pass RC network (contd.)

- Consider the behaviour of this circuit at different frequencies:

- **when $f \ll f_c$**

– $f/f_c \ll 1$, the voltage gain ≈ 1

- **when $f = f_c$**

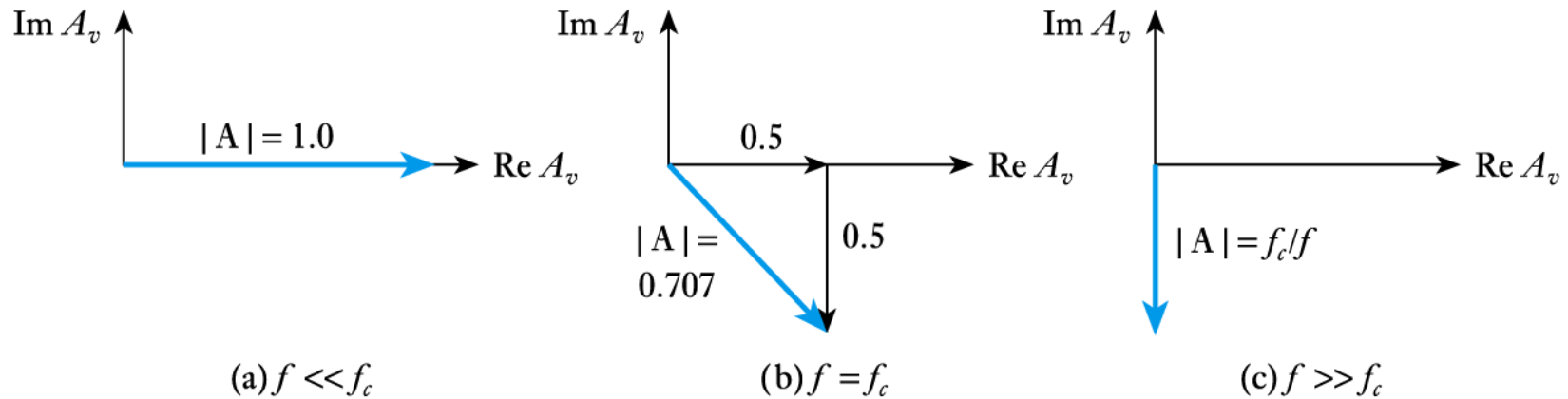
$$\frac{V_o}{V_i} = \frac{1}{1 + j\frac{f}{f_c}} = \frac{(1-j)(1+j)}{(1+j)} = \frac{(1-j)}{2} = 0.5 + 0.5j$$

- **when $f \gg f_c$**

$$\frac{V_o}{V_i} = \frac{1}{1 + j\frac{f}{f_c}} \approx \frac{1}{j\frac{f}{f_c}} = -j\frac{f_c}{f}$$

A low-pass RC network (contd.)

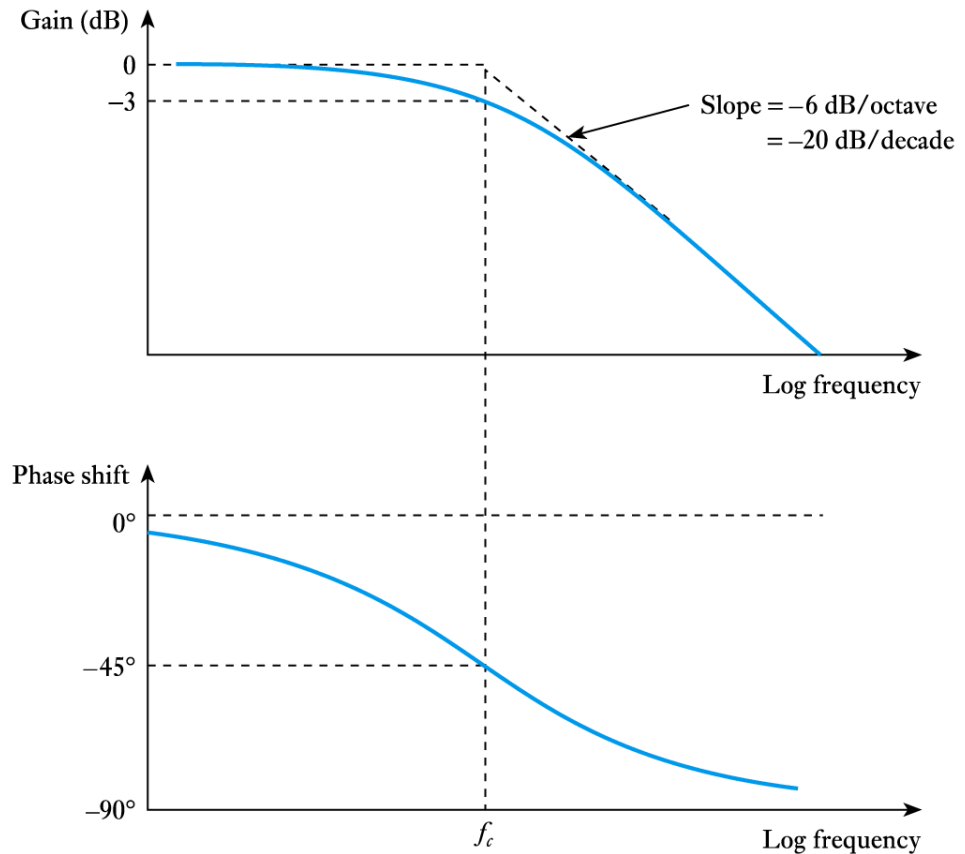
- The behaviour in these three regions can again be illustrated using phasor diagrams.



- At *high* frequencies the gain is linearly related to frequency. It falls at 6dB/octave (20dB/decade).

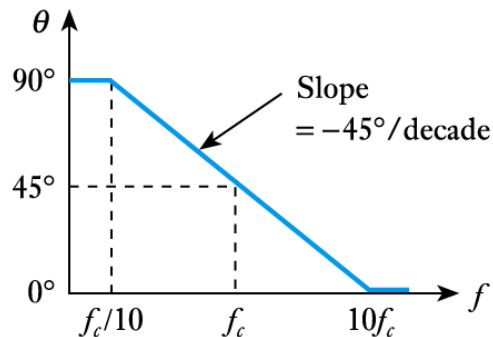
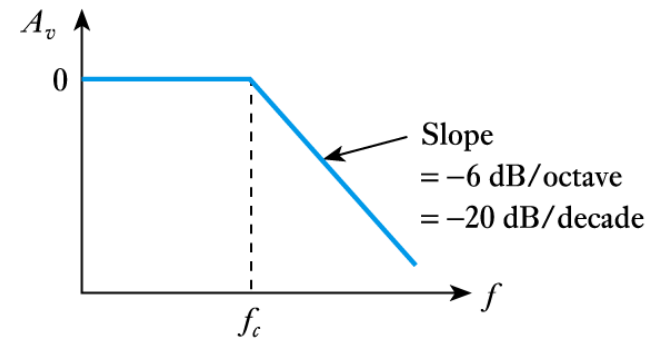
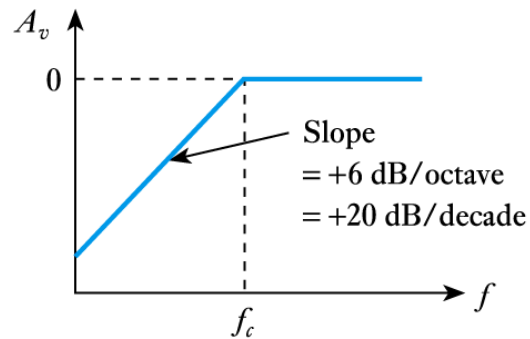
A low-pass *RC* network (contd.)

- Frequency response of the low-pass network
 - the gain response has two *asymptotes* that meet at the cut-off frequency.
 - you might like to compare this with the Bode Diagram for a high-pass network.

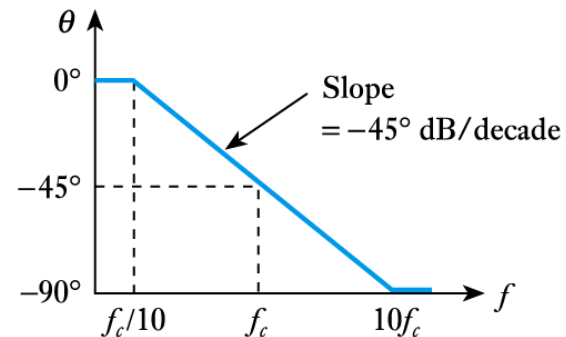


Bode diagrams

■ Straight-line approximations



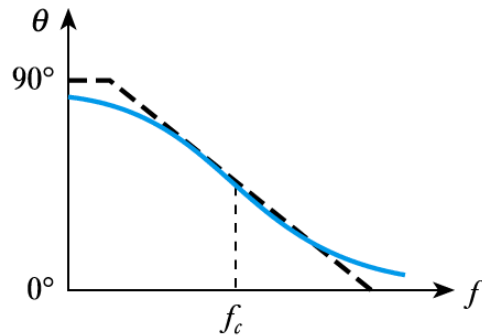
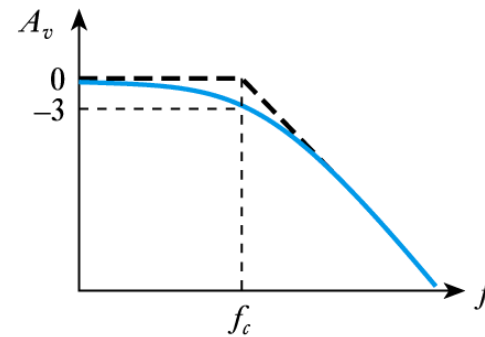
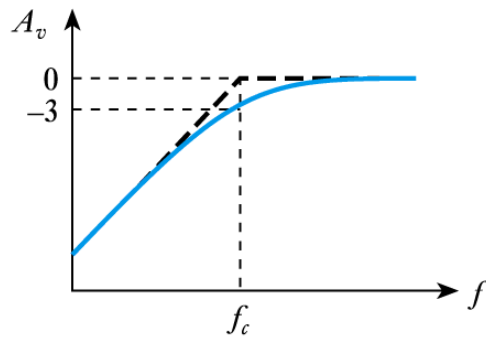
(a) High-pass circuit



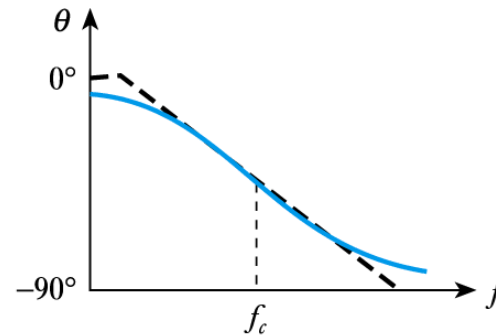
(b) Low-pass circuit

Bode diagrams (contd.)

- Creating more detailed Bode diagrams



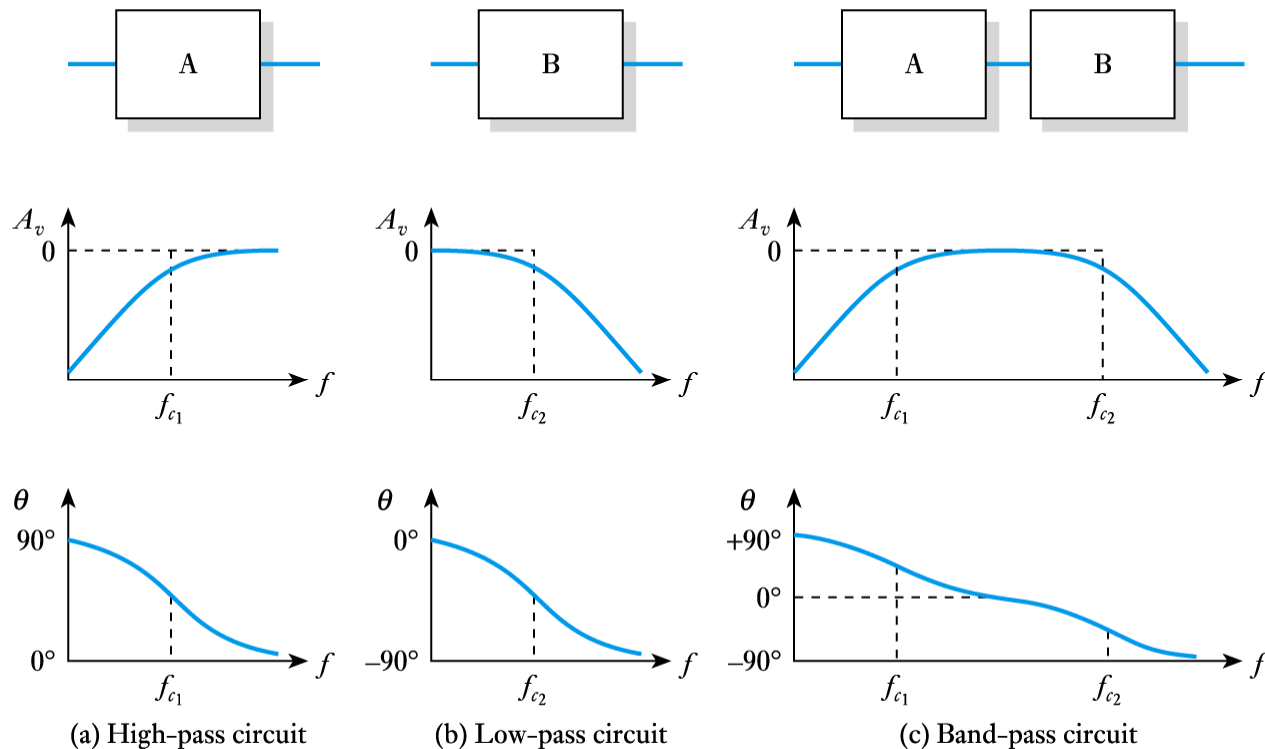
(a) High-pass circuit



(b) Low-pass circuit

Combining the effects of several stages

- The effects of several stages 'add' in Bode diagrams



Combining the effects of several stages (contd.)

- Multiple high- and low-pass elements may also be combined
 - this is illustrated in **Figure 3.25** in the course text.

